⭐ Windows Server User Security Hardening (User-Level)

Introduction

In this project, I implemented a series of **user-level security hardening configurations** within a Windows Server domain environment. The goal was to strengthen endpoint security, enforce compliance, reduce attack vectors, and control system usage for standard users.
These configurations were applied using **Group Policy Objects (GPOs)** and **Active Directory user properties**, ensuring centralized and consistent security enforcement.

Before applying user-level security hardening policies, ensure the following infrastructure and AD setup are completed:

Key User-Side Security Hardening Points

Server Deployment & Domain Setup

- **Deploy at least two virtual machines: one Windows Server and one Windows Client.**
- **Install Active Directory Domain Services (AD DS) on the Windows Server.**
- **Promote the server to a Domain Controller (DC) to manage domain policies.**
- **Join the Windows Client machine to the domain to ensure it can receive policies from the DC.**

Active Directory Structure

- **Create user accounts in Active Directory Users and Computers (ADUC) for all standard users.**
- **Design an organized OU structure, including at least:**
  - **Users OU**
  - **Machines/Endpoints OU**
  - **Move user accounts and machines into their respective OUs to allow targeted GPO application.**

✅ **These prerequisites ensure:**

- **The domain environment is ready for GPO-based security policies.**
- **Policies can be applied per-user and per-machine effectively.**
- **Endpoints and users are properly organized for centralized management.**

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1. Password & Authentication Security

- Enforced **strong password policies** for all users:
  - Minimum length
  - Complexity requirements
  - Password expiration and history
- Applied **Account Lockout Policy** to lock users after multiple failed login attempts.
- Increased account security and reduced the risk of unauthorized access.

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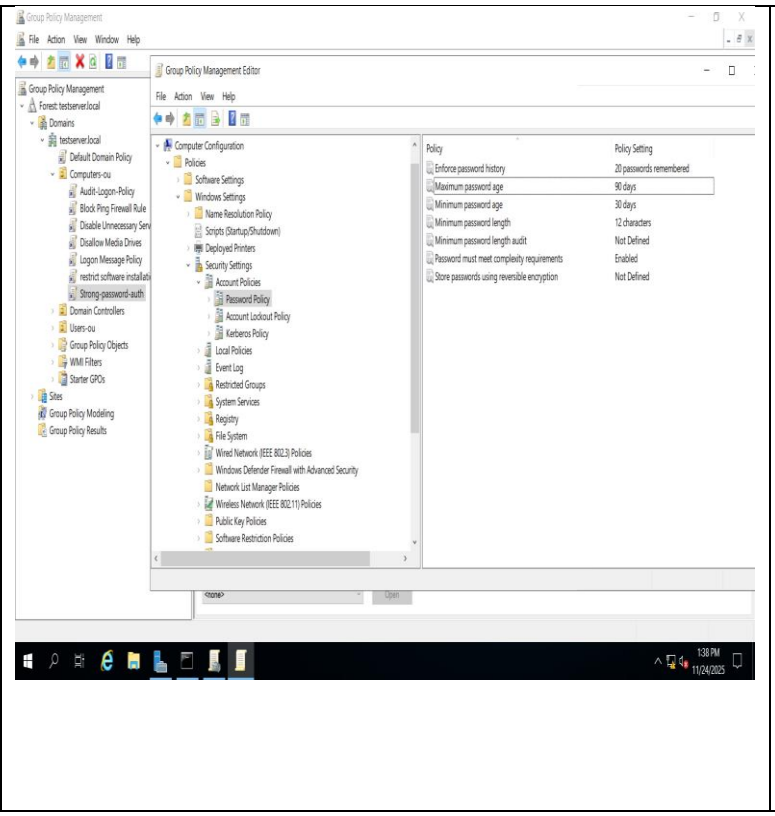

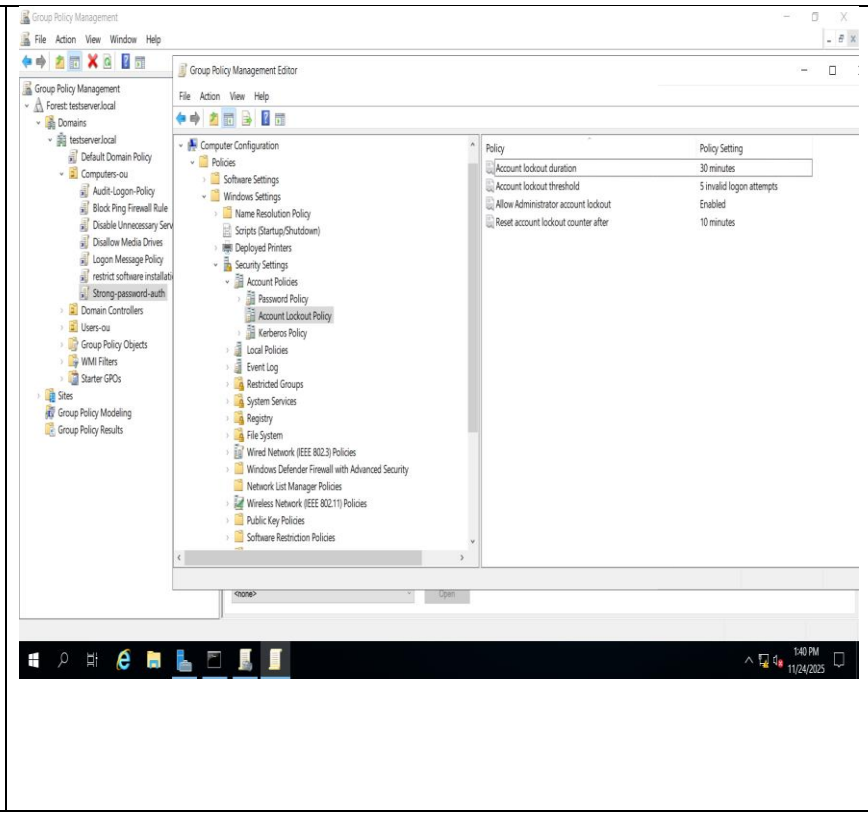

2. Enforced Screensaver Security (User-Level)

- Applied **user-scoped screensaver policy** to ensure automatic screen lock after inactivity.
- Enforced password-protected screen lock for improved physical security.
- Ensured all users follow the same session timeout standards.

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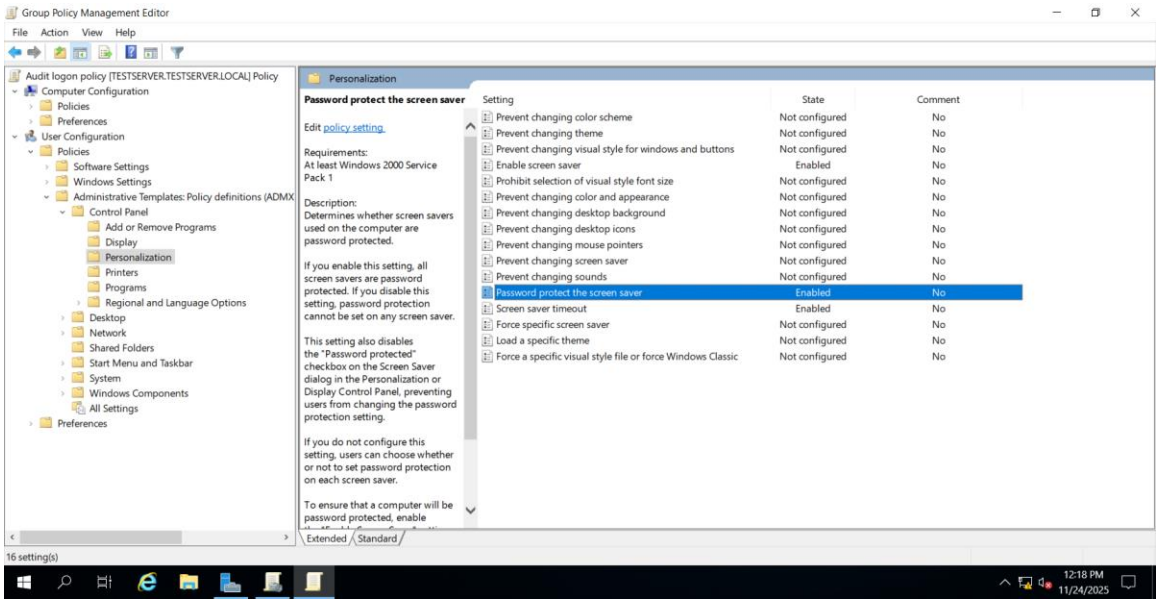

3. Implemented User-Side Logon Message

- Configured a **custom logon banner** shown before user authentication.
- Communicated acceptable use, security notices, or legal warnings.
- Increased user awareness of security policies.

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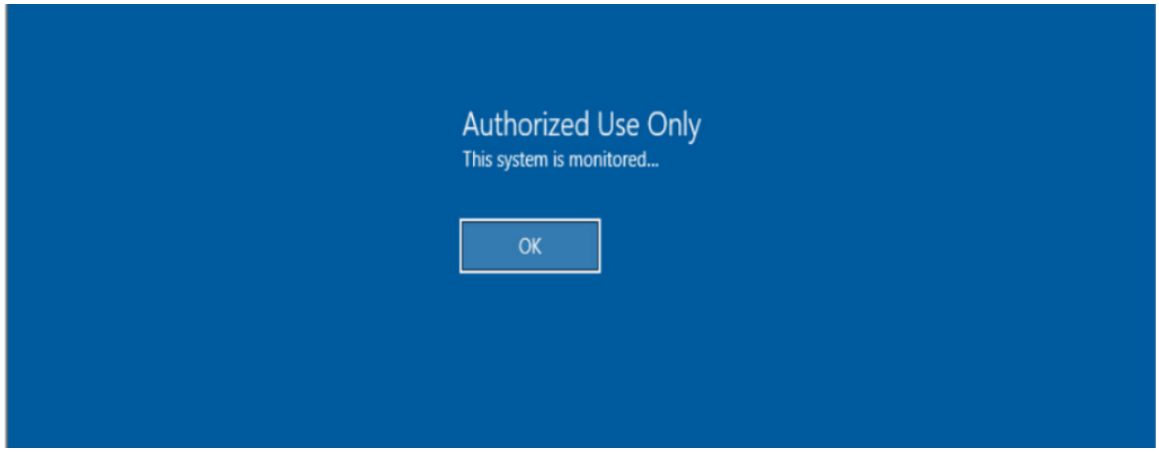

4. Disabled CMD & PowerShell (User-Scoped)

- Restricted standard users from accessing:
  - **Command Prompt**
  - **Windows PowerShell**
- Prevented execution of unauthorized scripts or commands.
- Reduced opportunities for privilege escalation and attack execution.

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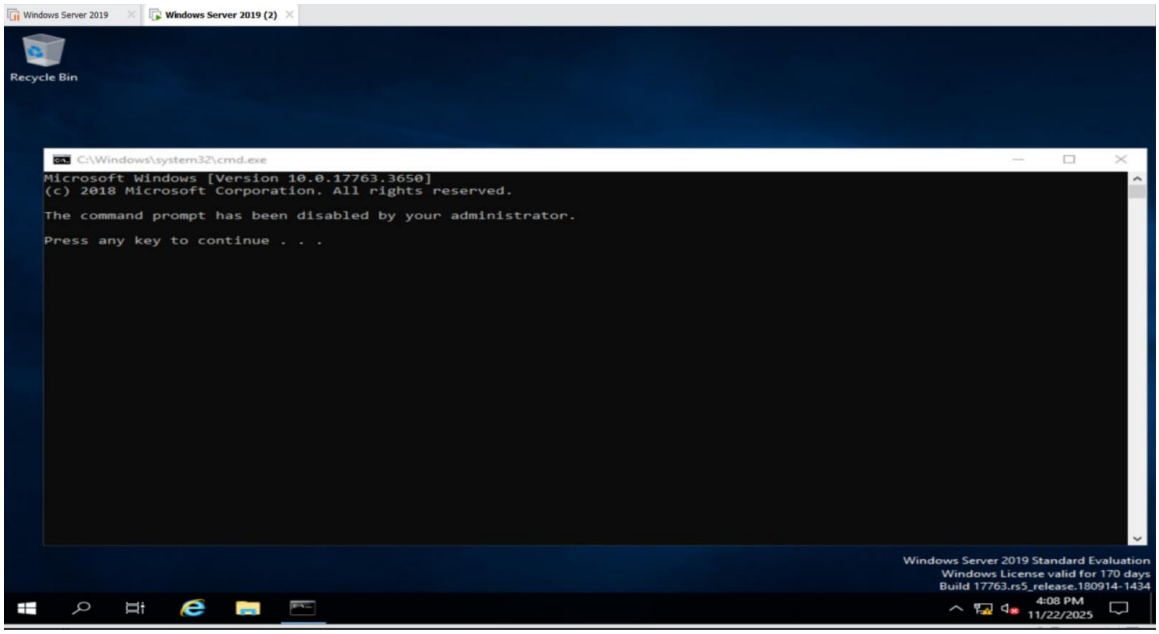

5. Disabled Access to Registry Editor (User Variant)

- Blocked access to **regedit.exe** for non-admin users.
- Prevented unauthorized modification of critical system registry settings.
- Maintained integrity and stability of endpoint configurations.

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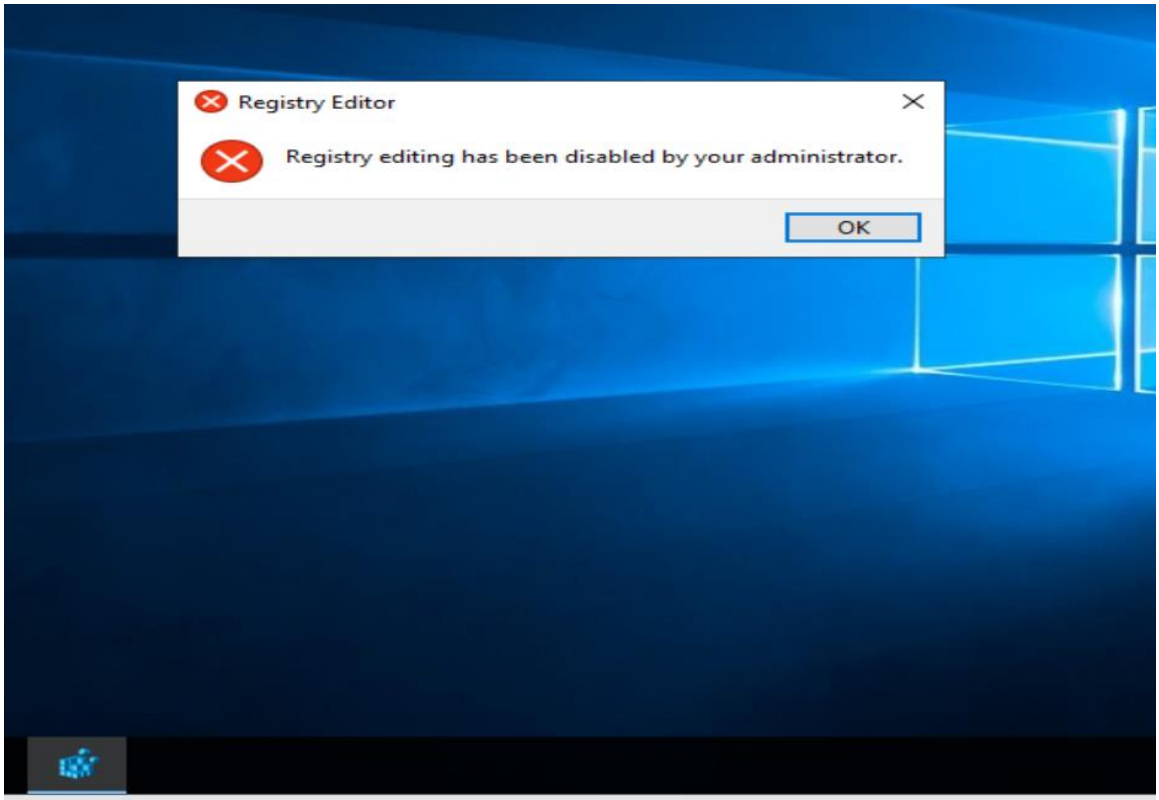

6. Restricted Time-of-Day Login (User Object Policy)

- Applied **logon hour restrictions** directly to user accounts.
- Allowed authentication only during approved work hours.
- Reduced unauthorized off-hours access attempts.

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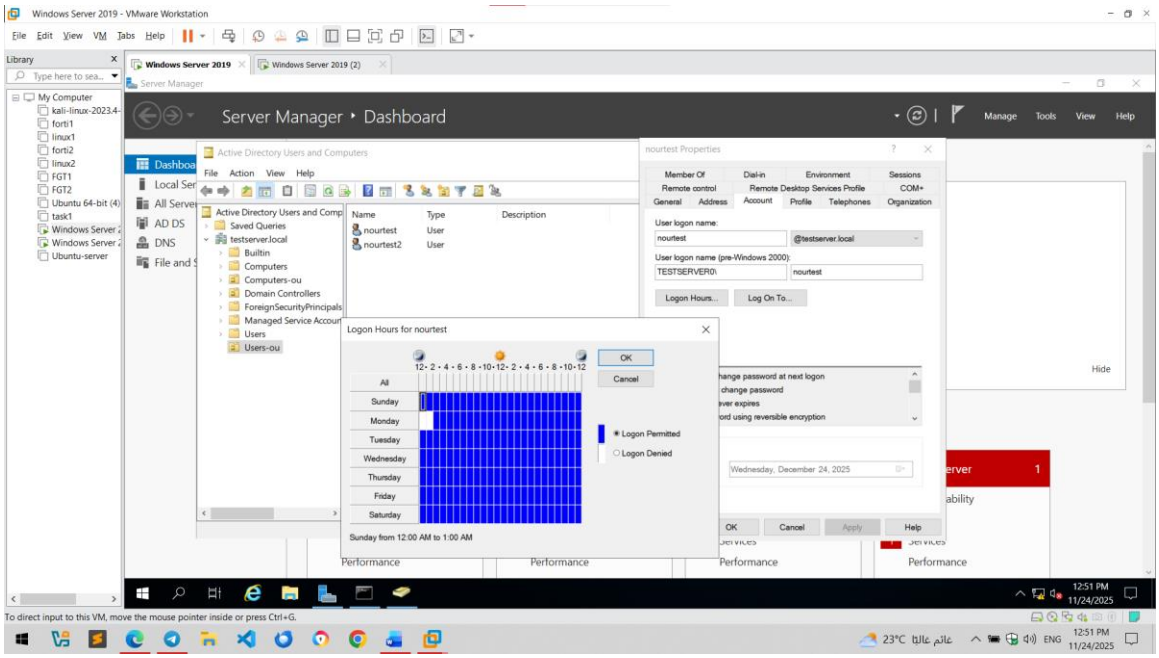

7. Applied Software Restriction Policies (Per User)

- Implemented **Software Restriction Policies (SRP)** to control which applications users can run.
- Blocked execution of untrusted or unknown software.
- Helped prevent malware, unauthorized tools, or unapproved scripts from running.

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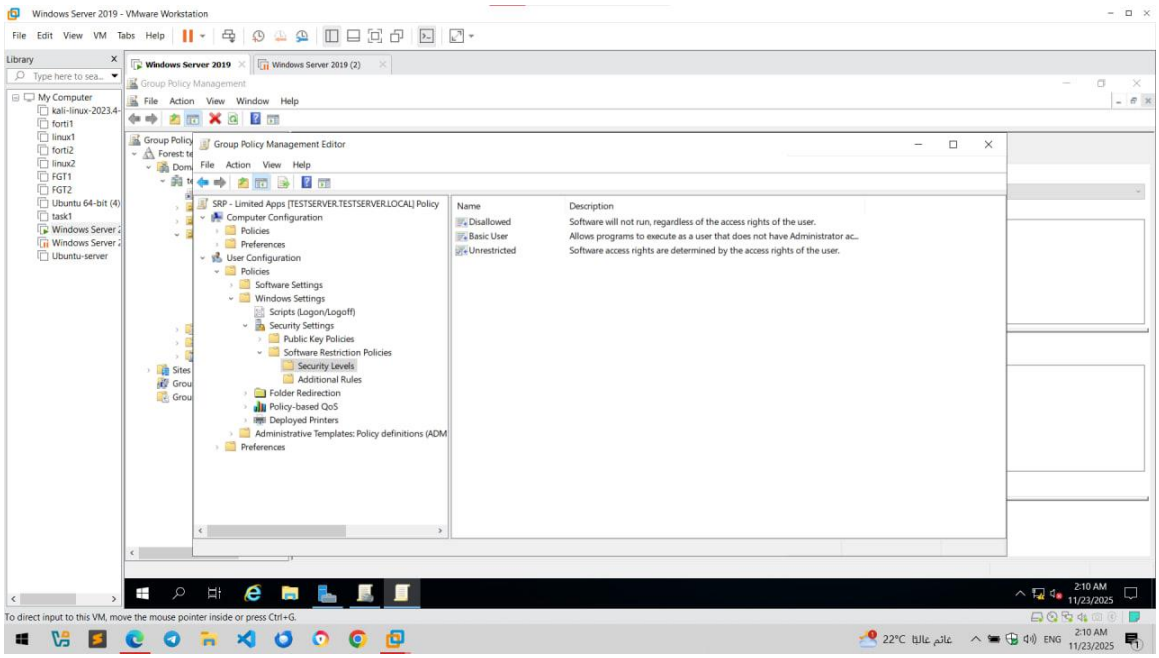

8.Disabled Services (Policy Level)

- Applied a GPO to **disable unnecessary or vulnerable Windows services** on endpoints.
- Reduced the attack surface and prevented misuse of system services.
- Ensured critical services remained operational while securing endpoints.

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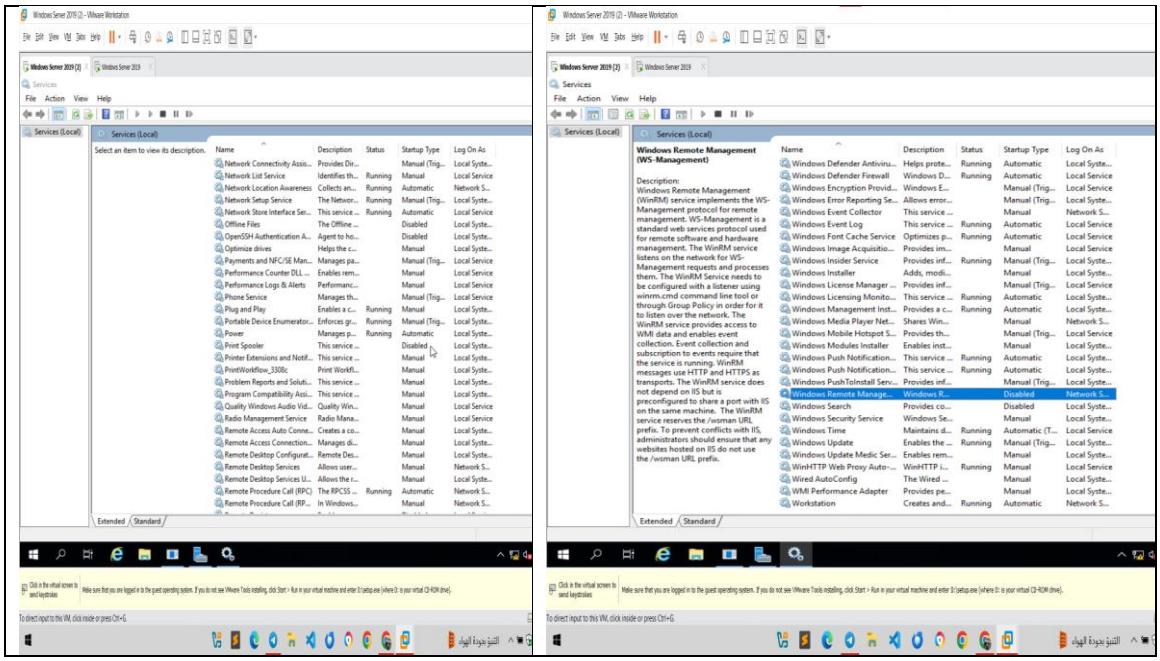

9. Firewall Rule – Block ICMP Ping (8.8.8.8 & 8.8.4.4)

- Configured a **domain-level firewall rule** to block ICMP ping requests to:
  - **8.8.8.8**
  - **8.8.4.4**
- Prevented unauthorized network probing and reduced external exposure.
- Applied via GPO to all domain-joined machines.

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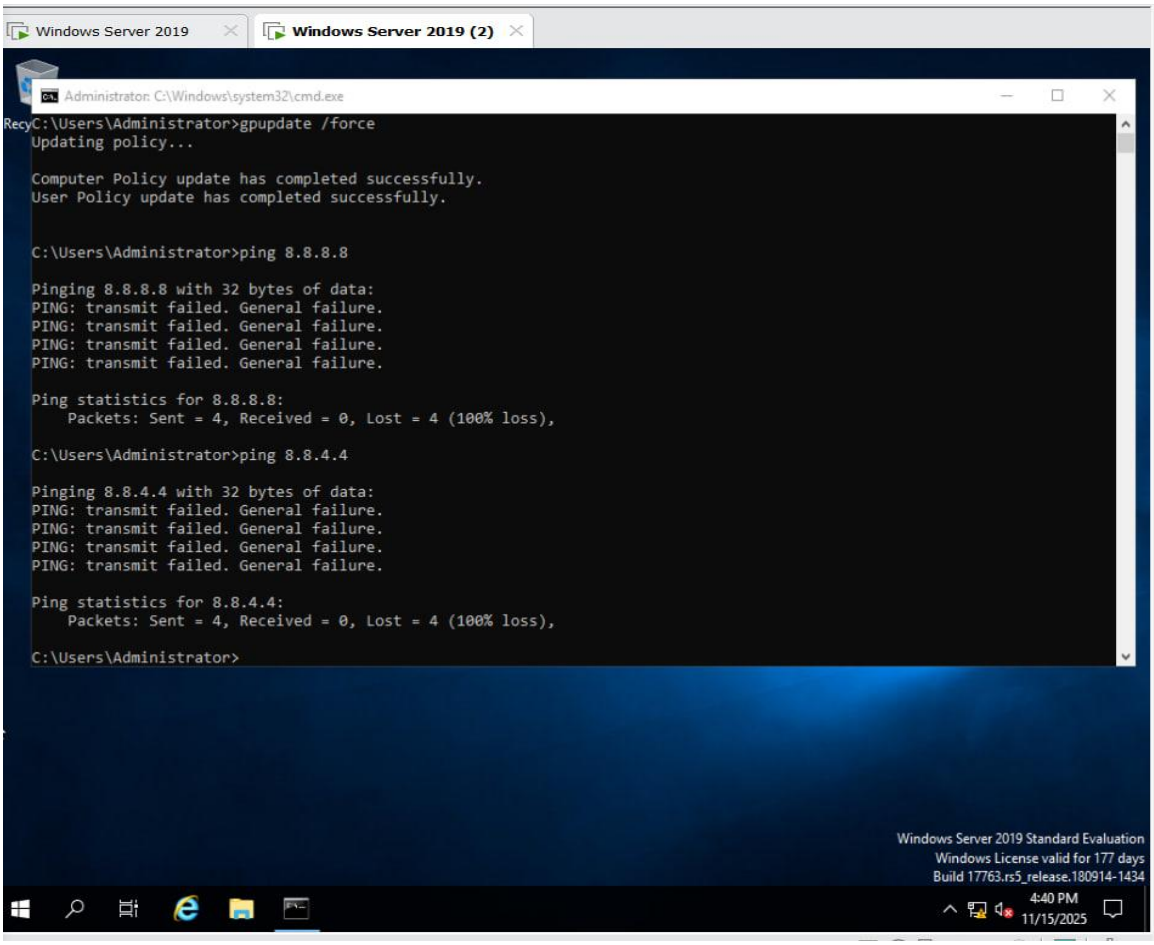

This PDF summarizes the **key user-level and endpoint security hardening measures** implemented in a Windows Server environment.
It demonstrates practical application of GPOs and AD configuration to **strengthen endpoint security**, ensure compliance, and enforce organizational policies….